Project Report: Author Detection Using NLP

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Project: Authorship Attribution (Who Wrote This?)

Dataset: 30Columnist Dataset

1. Executive Summary

This project aims to build a machine learning model capable of identifying the author of a given text based on their writing style and content. Using the **30Columnist dataset**, the project implements a Natural Language Processing (NLP) pipeline that involves text preprocessing, feature extraction using TF-IDF, and supervised classification. The analysis highlights not only the predictive capability of the model but also provides interpretability by extracting characteristic keywords for each author.

2. Approach and Experiments

2.1. Problem Definition

The task is a **multi-class classification problem** where the input is a text segment (column/article), and the output is one of the 30 specific authors found in the dataset.

2.2. Data Preprocessing

To convert raw text into a format suitable for machine learning, a rigorous preprocessing pipeline was applied:

- **Cleaning:** Removal of noise such as special characters, punctuation, and numerical values that do not contribute to authorship style.
- **Normalization:** Converting all text to lowercase to ensure consistency (e.g., "Football" and "football" are treated as the same feature).
- **Tokenization & Stopword Removal:** Breaking text into individual words and removing common language stop words (e.g., "the", "is", "and") to focus on meaningful content words.

2.3. Feature Engineering

The project utilized **TF-IDF (Term Frequency-Inverse Document Frequency)** for vectorization.

- **Why TF-IDF?** Unlike simple word counts, TF-IDF weighs words by how unique they are to a specific document compared to the entire corpus. This helps in identifying words that

are distinctively used by specific authors while downplaying generic terms.

2.4. Modeling

A supervised learning classifier was trained on the vectorized data. The model learned to map the statistical patterns of word usage (features) to specific author labels.

3. Model Results and Insights

3.1. Interpretability: Author Signatures

A key component of the results was the extraction of the "Top N" most important features (words) for each author. This analysis reveals that the model effectively captured the **topical focus** of the authors, which serves as a strong proxy for identity in this dataset.

Selected Author Profiles (from Model Analysis):

- **The Sports Commentator (Author 28):**
  - *Keywords:* club, league, arsenal, football, premier, referees, chelsea.
  - *Insight:* This author is clearly a sports journalist, specifically focused on the English Premier League. The model relies on specific team names and football terminology to detect them.
- **The US Political Analyst (Author 5):**
  - *Keywords:* mccain, obama, hillary, palin, senate, clinton, presidential.
  - *Insight:* This author writes heavily about US elections and political figures. The presence of specific names ("Hillary", "McCain") suggests the dataset may be from a specific historical window (likely around the 2008 US election)..
- **The Science/Health Writer (Author 29 & 9):**
  - *Keywords (Auth 9):* cancer, research, cells, drug, disease, treatment.
  - *Keywords (Auth 29):* studies, scientists, alcohol, skin, symptoms.
  - *Insight:* These authors are distinguished by technical medical vocabulary. The model separates general science reporting from specific medical research discussion.
- **The Financial Analyst (Author 30):**
  - *Keywords:* percent, billion, million, gasoline, trillion, wall street.
  - *Insight:* Quantitative terms and financial entities ("Wall Street") are the primary signals for this class.

3.2. Conclusion

The experiments demonstrate that **Topic Modeling** is a dominant factor in authorship attribution for this specific dataset. The model successfully distinguishes authors not just by sentence structure, but by the specific *domain* they cover (Sports vs. Politics vs. Science). The TF-IDF approach was highly effective in isolating these domain-specific keywords, providing high interpretability for *why* a specific text was classified to a specific author.